

Steven Counterman

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TARGET ROLES

Systems / Platform SWE • Developer Tools / DevEx • SRE / Production Engineer • Security Engineering

SUMMARY

Computer Science graduate (B.S., May 2026) with a strong systems foundation (OS, architecture, networking/security) and hands-on experience building developer-facing tooling and execution visualizations. Research accepted, presented, and published at PACISE 2026 on RosettASM, a pedagogical system for visualizing program execution. Army infantry veteran known for discipline, reliability, and calm execution in high-stakes environments.

EDUCATION

DeSales University – B.S. Computer Science | GPA: 3.99/4.00

Relevant coursework: Data Structures & Algorithms, Operating Systems, Computer Architecture, Networking & Security, Database Systems, Software Engineering, Programming Languages / Compiler Design (Independent Study).

TECHNICAL SKILLS

- Languages: Python, Java, C++, SQL, x86 Assembly (32/64-bit)
- Systems: Program execution & memory models, OS fundamentals, computer architecture, networking concepts
- Tools/Environments: Linux, Git, Eclipse, Oracle, Google Colab, ASP.NET Core, PostgreSQL, Apache (Reverse Proxy), Docker (Compose)

RESEARCH & PUBLICATION

- RosettASM: A Pedagogical System for Visualizing Program Execution – Accepted, presented, and published, PACISE Conference (2026)

PROJECTS

- RosettASM – Interactive systems – visualization toolchain that helps learners build accurate mental models of machine-level execution.
 - Built full compiler pipeline targeting x86-32 assembly
 - Designed mapping between high-level source code and generated assembly for step-by-step analysis
 - Built visualizations of registers, memory, stack, and control flow to make runtime state observable for learning and debugging
 - Selected for integration into university coursework for instructional use at DeSales University
- Code Jam – Team Lead / Systems Integrator
 - Led senior project team building a full-stack programming competition platform
 - Integrated ASP.NET frontend, Python execution service, and PostgreSQL database into a multi-service architecture
 - Deployed to Linux using Apache reverse proxy; resolved cross-service integration issues
- Project Workflow Manager – Agile Project Management System
 - Developed application to manage projects, phases, and team assignments using structured workflows
 - Applied Agile/Scrum practices including sprint planning, task tracking, and iterative development
 - Implemented features for project phase management, scheduling, and resource allocation
 - Selected for university course use; continuing development for deployment in Software Engineering and Senior Coordinating Seminar courses

EXPERIENCE

- United States Army – Infantry (2020-2021): Operated in high-stress, team-based environments requiring strict procedural compliance; demonstrated reliability and accountability in mission-critical situations.